WakeBand flex rev A: 2-layer polyimide flex, folds once around a 12 x 30 mm 110 mAh cell.
Pod about 22 x 36 x 8 mm. Sheets connect through global labels and power nets.
Status: steps 1 and 2 done (project, ISP2454 symbol, GPIO plan); step 3 power sheet (nPM1304) drafted; other sheets are notes.

power

Power: nPM1304 PMIC, battery, NTC, ship mode, current jumper

File: power.kicad_sch

mcu

MCU: ISP2454 (nRF54L15) module, SWD, UART, I2C, interrupts

File: mcu.kicad_sch

sensors

Sensors: LIS2DW12 accelerometer

File: sensors.kicad_sch

haptics

Haptics: DRV2605L driver and LRA pads

File: haptics.kicad_sch

dock

Dock: six pogo pads, TVS, series resistors

File: dock.kicad_sch

debug

Debug: test pads, debug LED (DNP option)

File: debug.kicad_sch

Plan: docs/flex-pcb-schematic-plan.md

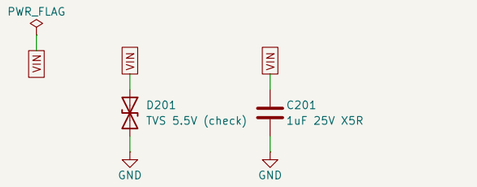
**FOSS Health Alarm**

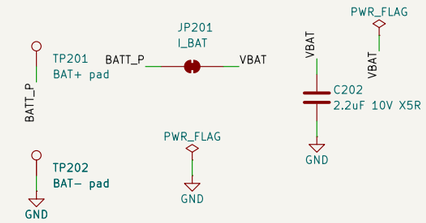
Sheet: /
File: flex-rev-a.kicad_sch

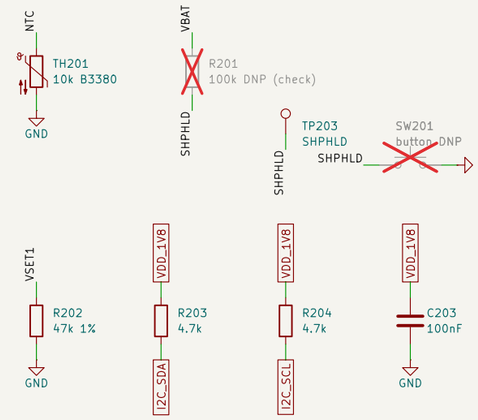
***Title: WakeBand flex rev A — Top***

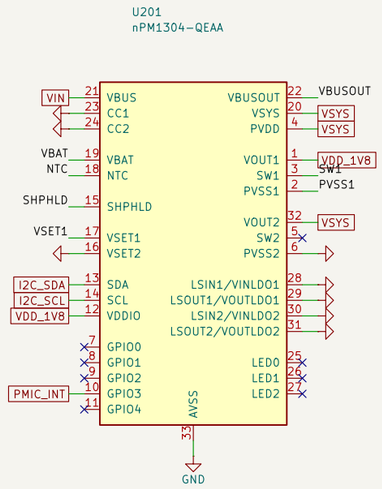
| Size: A4 | Date: 2026-10-30 | **Rev: A0** |
|---|---|---|
| KiCad E.D.A. 10.0.5 | | Id: 1/7 |

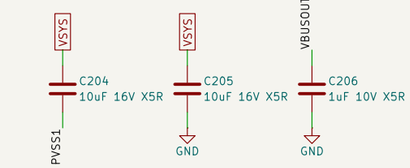
nPM1304 power sheet (step 3). Based on Nordic nPM1304 datasheet, reference circuit 'Configuration 2' (QFN):
one buck, no LDO/load switch, ship-mode exit and NTC used. Draft until the Stage 0b nPM1304-EK tests.

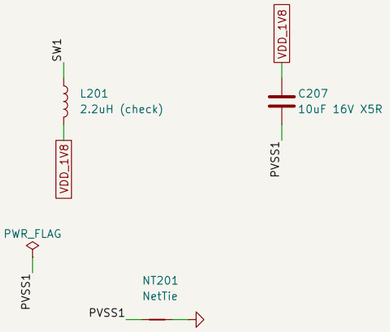



Strapping and connections (from the datasheet):
- VBUS from dock pad VIN. VBUS abs max 22 V; OVP 5.5 V typ; VBUSPOR 3.9 V. TVS: Vrwm >= 5.5 V, clamp < 22 V.
- CC1/CC2 to GND: no USB-C; VBUS limit stays at the 100 mA default (enough for 40-55 mA charging).
- VBUSOUT not used by the SoC but needs its 1 uF decoupling cap.
- VSET1 = 47k -> BUCK1 starts at 1.8 V (datasheet table: 47k = 1.8 V, max 5% tolerance). VOUT1 = VDD_1V8.
- BUCK2 unused: VSET2 to GND (off), VOUT2 to VSYS, SW2 open, PVSS2 to GND, no inductor (as in reference config 2).
- LOADSW1/2 unused: LSIN1, LSOUT1, LSIN2, LSOUT2 to GND (as in reference config 3). LED0-2 open.
- VDDIO from VOUT1. Unused GPIO0/1/2/4 open: default is input with weak pull-down.
- PMIC_INT on GPIO3 (any GPIO can be the interrupt; set in devicetree).
- PVSS1 joins GND through net tie NT201 (one via to the GND layer), as in the reference layout.
- NTC: 10k 1% B25/50 = 3380 K, one of the three supported types; JEITA defaults match it. Max 100 pF across it.
- SHPHLD has an internal 50k pull-up. Ship mode exits on SHPHLD low (button to GND) OR VBUS > VBUSPOR, so docking alone wakes the band (still confirm on the EK in Stage 0b). R201 is an optional extra pull-up to VBAT.
- TWI address 0x6B. I2C pull-ups R203/R204 serve the whole 1.8 V bus (4.7k for 400 kHz).
- JP201: cut to put the PPK2 in series with the cell; bridge again with solder.

(check) before freezing:
- U201 footprint: KiCad QFN-32 5x5 EP 3.45 mm; datasheet EP is 3.5 nom (3.4-3.6). Compare with Nordic's QFN reference layout.
- D201 part number (leakage < 50 nA at 5 V). L201 part number, R201 value (or delete).
- GPIO3 for PMIC_INT: match the nPM1304-EK / Zephyr devicetree used in Stage 0b.
- BUCK1 at 1.8 V -5% = 1.71 V vs the nRF54L15 1.7 V minimum.
- VSYS abs max 5.5 V: dock supply must stay at 5.0 V; DRV2605L blocked while VBUS is present.

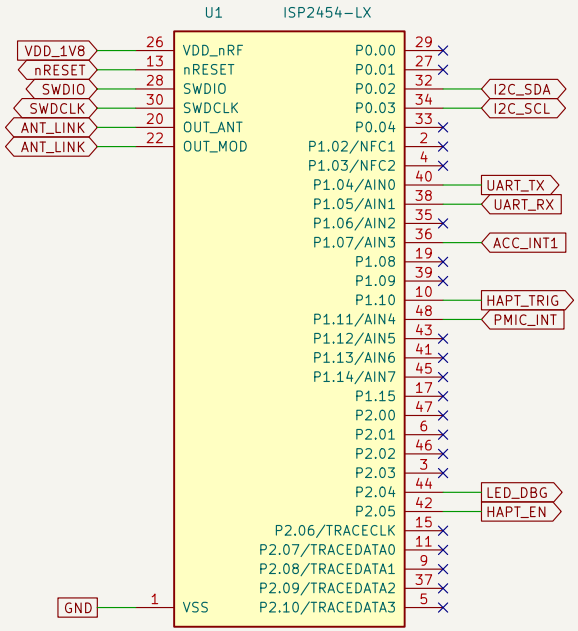
nPM1304 per Nordic reference circuit configuration 2 (QFN32)
Plan: docs/flex-pcb-schematic-plan.md
**FOSS Health Alarm**

Sheet: /power/
File: power.kicad_sch

***Title: WakeBand flex rev A — Power***

| Size: A3 | Date: 2026-10-30 | **Rev: A0** |
|---|---|---|
| KiCad E.D.A. 10.0.5 | | Id: 2/7 |

ISP2454 notes (datasheet R7):
- Antenna, 32 MHz + 32.768 kHz crystals, DC-DC inductor and all decoupling are inside. No external parts.
- OUT_ANT (20) must be joined to OUT_MOD (22) by a short trace (net ANT_LINK).
- Pads 49-62 are mechanical: isolated copper in the footprint, not in this symbol.
- Antenna corner 1 mm from the board edge, no copper or parts on any layer in the keep-out, including the folded battery layer below it.
- Firmware: set LFXO load capacitance to 19 pF (devicetree load-capacitance-femtofarad = <19000>).
- Every signal uses an OUTER-row pad, so the flex needs no via-in-pad. Inner GND pads (21, 23) and nRESET (13) may use tented vias under the module centre, as the datasheet allows.
- I2C on TWIM30 (P0, low-power domain). UART on UARTE20 (P1).
- P1.02/P1.03 are NFC pins: left unused. P2 trace pins left unused.
- GPIO plan and reasons: docs/flex-rev-a-pinout.md




Plan: docs/flex-pcb-schematic-plan.md

**FOSS Health Alarm**

Sheet: /mcu/

File: mcu.kicad_sch

***Title: WakeBand flex rev A — MCU***

| Size: A4 | Date: 2026-10-30 | **Rev: A0** |
|---|---|---|
| KiCad E.D.A. 10.0.5 | | Id: 3/7 |

Sensors sheet (step 5): LIS2DW12 LGA-12 on VDD_1V8.
- CS tied to VDD_1V8 (I2C mode), SA0 strapped high -> address 0x19.
- SDA/SCL on I2C_SDA / I2C_SCL, 4.7 k pull-ups to VDD_1V8 (shared bus).
- INT1 -> ACC_INT1 (wake), INT2 -> ACC_INT2.
- 100 nF on VDD and VDD_IO.

Plan: docs/flex-pcb-schematic-plan.md
**FOSS Health Alarm**
Sheet: /sensors/
File: sensors.kicad_sch
***Title: WakeBand flex rev A — Sensors***
Size: A4 | Date: 2026-10-30 | **Rev: A0**
KiCad E.D.A. 10.0.5 | Id: 4/7

Haptics sheet (step 5): DRV2605L VSSOP-10 (DRV2605LDGS) on VSYS.
- 1 uF on VDD and on REG. EN <- HAPT_EN, IN/TRIG <- HAPT_TRIG.
- I2C at 1.8 V (VIH 1.3 V), address 0x5A. Keep VDD powered (an unpowered driver clamps the bus).
- OUT+/OUT- to two LRA lead pads beside the stiffened 9 mm LRA tape-down pad.
- Firmware blocks the motor while docked (VSYS can reach 5.5 V).

Plan: docs/flex-pcb-schematic-plan.md
**FOSS Health Alarm**
Sheet: /haptics/
File: haptics.kicad_sch
***Title: WakeBand flex rev A — Haptics***
Size: A4 | Date: 2026-10-30 | **Rev: A0**
KiCad E.D.A. 10.0.5 | Id: 5/7

Dock sheet (step 6): six pads at 2 mm pitch, about 1.2 x 3 mm:
1 VIN  2 GND  3 SWDCLK  4 SWDIO  5 UART_RX  6 UART_TX
- TVS on every pad, leakage under 50 nA; VIN TVS working voltage at least 5.5 V.
- 100 ohm in series on the four data pads. Pull-down on UART_RX.

Plan: docs/flex-pcb-schematic-plan.md
**FOSS Health Alarm**
Sheet: /dock/
File: dock.kicad_sch
***Title: WakeBand flex rev A - Dock***
Size: A4 | Date: 2026-10-30 | **Rev: A0**
KiCad E.D.A. 10.0.5 | Id: 6/7

Debug sheet (step 6): test pads for every signal plus nRESET, SHPHLD, ACC_INT2 and spare GPIOs; one 0402 LED on LED_DBG with a DNP option.

Plan: docs/flex-pcb-schematic-plan.md

**FOSS Health Alarm**

Sheet: /debug/

File: debug.kicad_sch

***Title: WakeBand flex rev A — Debug***

| Size: A4 | Date: 2026-10-30 | **Rev: A0** |
|---|---|---|
| KiCad E.D.A. 10.0.5 | | Id: 7/7 |